AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 14 · STUDENT LESSON

Scores and Calibration Intuition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Name the event. Count the positives. Compare the group.

Interpret probability calibration without confusing it with chosen-label accuracy or individual certainty.

Keep one habit: Score contract → two-group calculation → pictorial contrast → complete thirty-card lab → independent new groups.

MAIN LESSON · SECTION 1

What event is this score trying to predict?

> Interpret calibration only after stating what the probability refers to.

Learning goals

I can interpret **calibration** using groups of forecasts.

I can compare average predicted probability with **observed frequency**.

I can distinguish **calibration** from individual correctness.

Recall prediction, fitting and score meaning

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a **score** for your ability.
- Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.
- Does a **score** necessarily represent a probability?

End goal

By the end, I can explain **score**s and **calibration** intuition using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A model **score** can rank **prediction**s without matching the real frequency of correct outcomes.



LEARNING TARGET Twenty cases receive **score** .7 and fourteen are correct. Explain what the observed rate suggests and what it does not establish about another **score** group.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Declare the score's event before making groups

> The main lesson uses probability of the positive class, not probability of any chosen label being right.

Score-event contract

Current score meaning	Correct comparison
p(positive), here p(spotted under a toy label rule)	Compare mean predicted p with fraction of reference-positive cases in that score group.
p(chosen label is correct), a separate confidence-calibration task	Compare with fraction of selected predictions that were correct; do not substitute this outcome into the positive-class example.
Arbitrary model score or neighbour vote share	Do not call it a probability without a justified interpretation and evaluation.
Evidence limit	Agreement in a small group is an observation, not a guarantee for an individual, another score bin or a new population.

Read the group interpretation and fictional scope

If many comparable cases receive probability .8, a calibrated model should produce positives in roughly 80 percent of those cases over adequate data.

It does not mean every case is 80 percent correct or that exactly eight of each next ten must be positive. **Calibration** is assessed across groups.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **calibration** as a group property.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

PICTURE ENTRY
</details>

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use **score**s and **calibration** intuition to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Compare average probability with observed positive frequency

> Keep score, outcome count and denominator visible.

Complete two-group supplied example

In a fictional held-out check, ten cases all have predicted probability .2 and two are positive. Another ten all have probability .8 and six are positive.

Average **prediction**s are .2 and .8; observed frequencies are 2/10=.2 and 6/10=.6. The second group is overconfident for positives in this sample.

Optional complete calculation script

```
groups = [(0.2, 10, 2), (0.8, 10, 6)]
for score, count, positives in groups:
    if count == 0:
        print("No evidence in this group")
    else:
        print(score, positives / count)
# Compare each score with its observed fraction.
# A group fraction is not an individual guarantee.
```

Predict the numbers before running

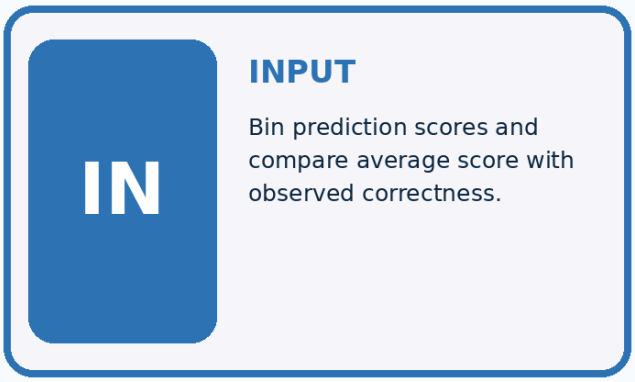
Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

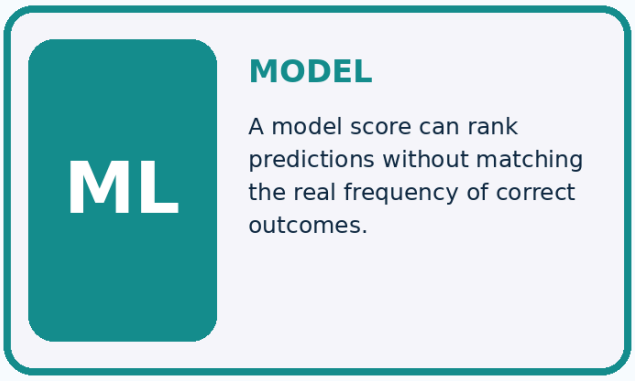
▼ Earlier wording, alternatives and picture transcripts

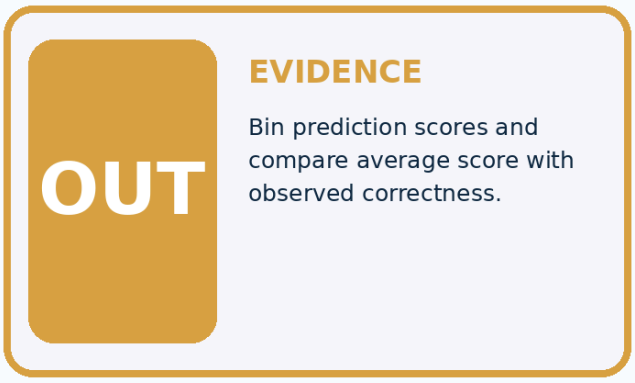
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Bin **prediction score**s and compare average **score** with observed correctness.

MODEL IDEA A model **score** can rank **prediction**s without matching the real frequency of correct outcomes.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD For this example, each **score** estimates the probability that the selected label is correct. Ten **prediction**s have **score**s near 0.8; six are correct, so observed correctness is 6/10=**60%**.

If instead **score**s mean probability of the spotted class, compare them with the fraction of reference-spotted leaves.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

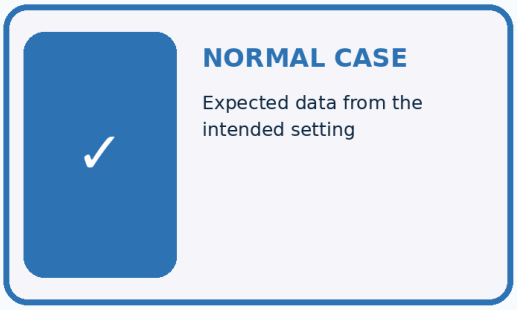
Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

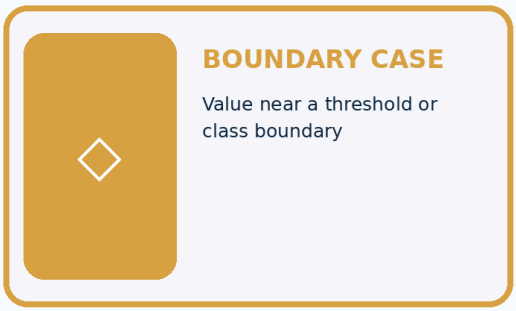
Compare a probability statement with the evidence for its meaning

> A familiar decimal is not automatically a reliable individual guarantee.

Four evaluation conditions



NORMAL Expected data and expected behavior.

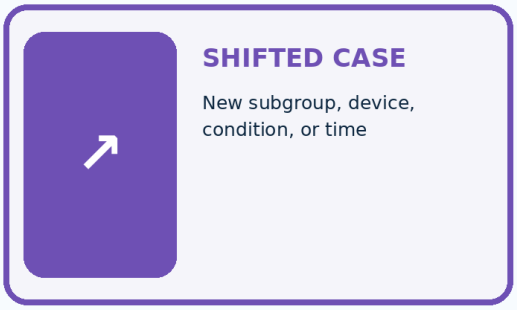


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: The 0.2 group has two spotted cases instead of four.

▼ **Check after predicting**

Prediction: Its observed fraction becomes 0.2 and matches the supplied **score** in this group.

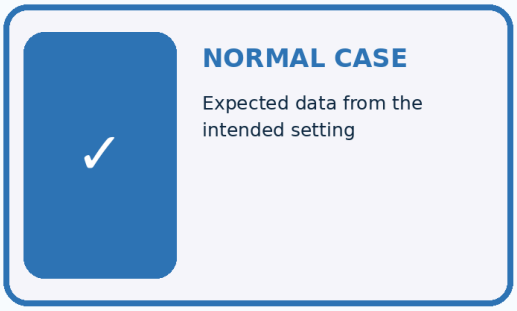
Reason: **Calibration** compares stated probabilities with group outcome frequencies.

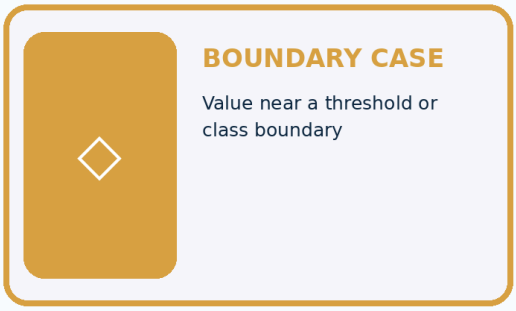
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

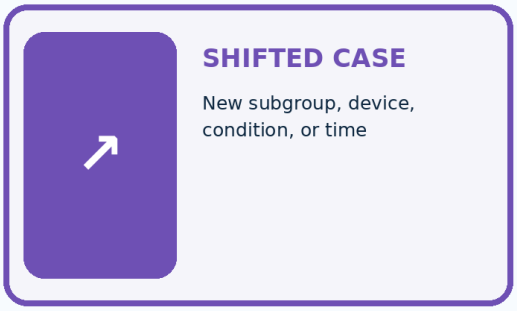
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read both axes and the event behind each count

> The positive fraction and classification accuracy can be different.

Complete pictured groups and separate-fixture scope

Treat **calibration** as a comparison across groups of probability **score**s.

Toy classifier **output**s p(spotted)=0.8 on ten cases; eight are actually spotted. Another ten have p=0.2; four are spotted.

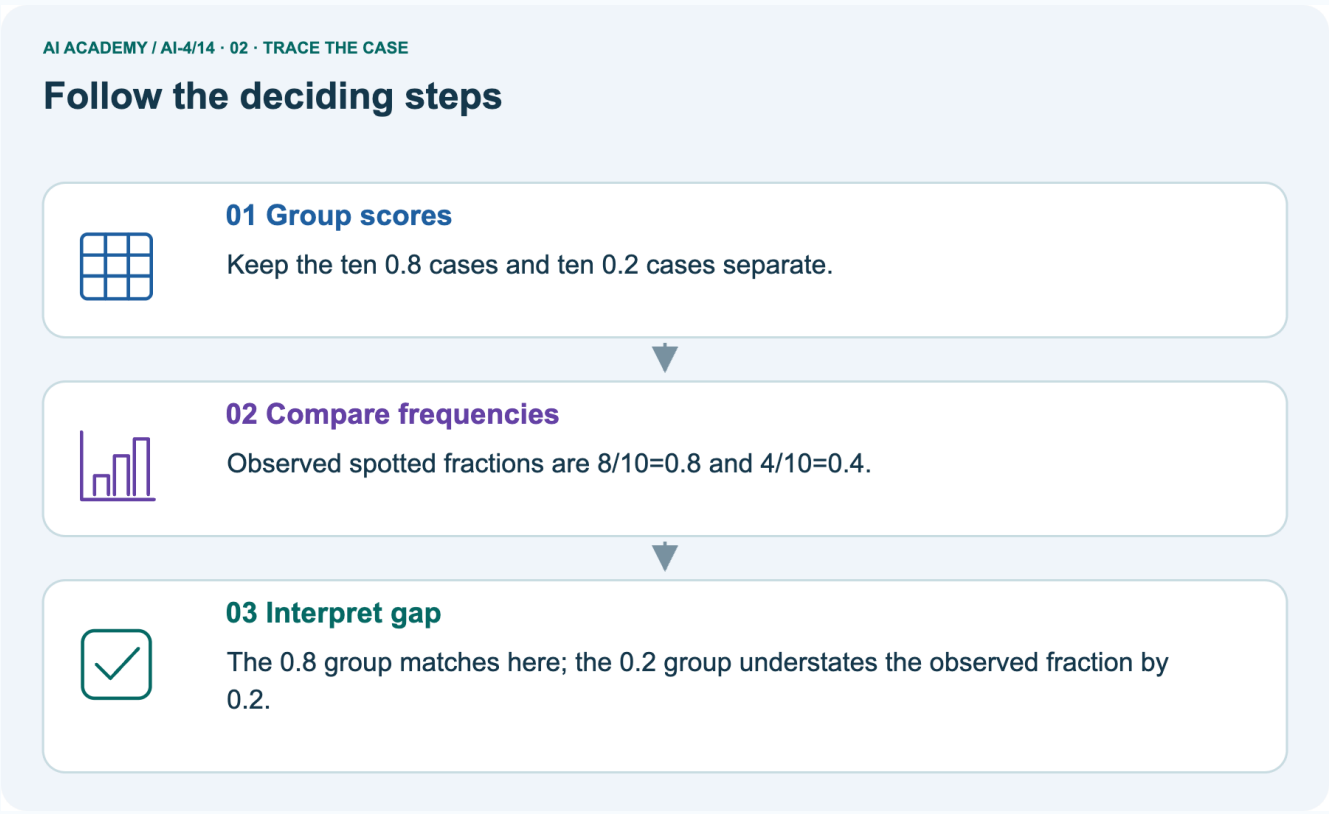
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The main calculation and picture are different fixtures. Main low group: p=0.2,2positives/10,point(0.2,0.2); main high: p=0.8,6/10,point(0.8,0.6),overstates by0.2 or20percentage points. Picture high: p=0.8,8/10; picture low: p=0.2,4/10,understates by0.2.

For a reliability diagram x is mean predicted positive probability and y observed positive fraction; compare with y=x.

Neither example tells whether every chosen class label is correct. Ten cases per group remain noisy evidence.

Trace positive frequencies and gaps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Group scores	Keep the ten 0.8 cases and ten 0.2 cases separate.
2. Compare frequencies	Observed spotted fractions are 8/10=0.8 and 4/10=0.4.
3. Interpret gap	The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

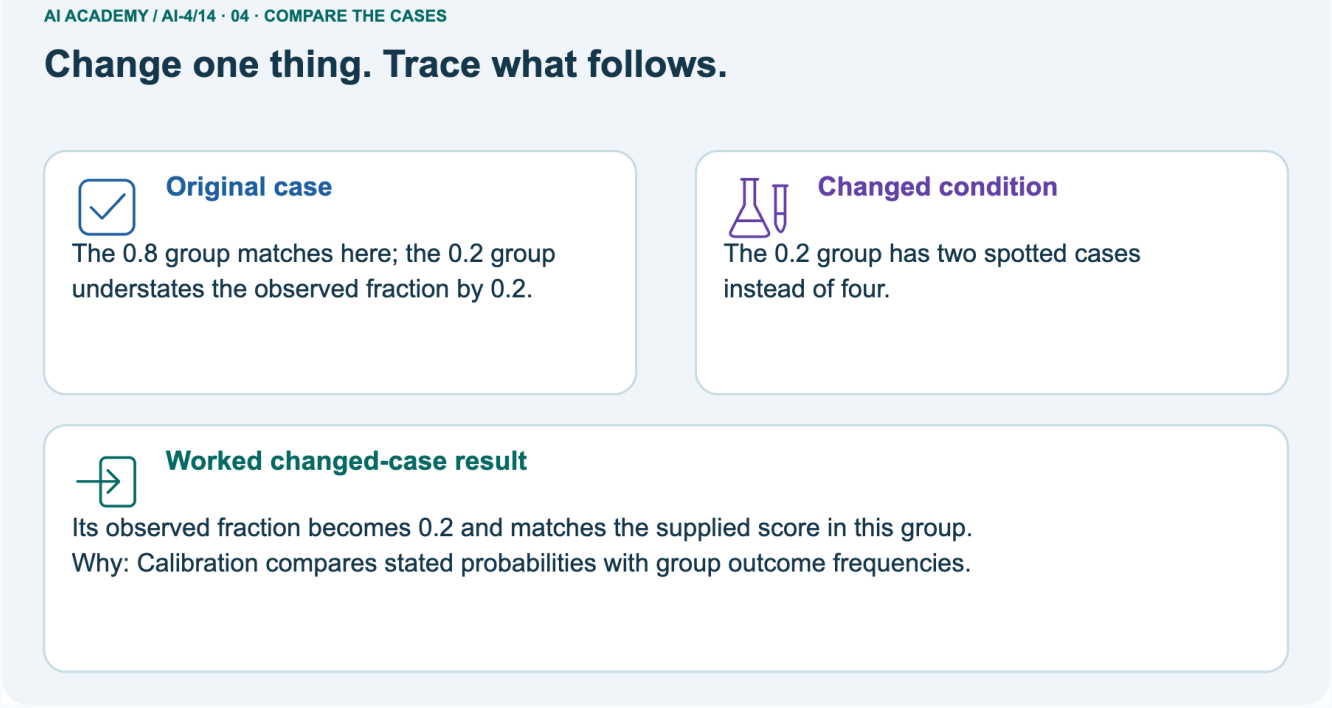
Read the picture in order. The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict changing the pictured low-group positives

- Change one thing: The 0.2 group has two spotted cases instead of four. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain individual and sample limits

- Ten cases per group provide noisy **evidence**; the exercise does not establish **calibration** on every **score** or setting.

- Read the labelled picture: Does a calibrated 0.8 **score** mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the original two-group calculation

Calibration as a group property at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Low probability group	Average .2	Observed 2 of 10
High probability group	Average .8	Observed 6 of 10
Compare groups	Prediction versus frequency	Keep group sizes visible

Plot average prediction on the horizontal axis and observed frequency on the vertical axis. The two points are (.2,.2) and (.8,.6). Agreement lies along the diagonal. Use the average score within a bin, not automatically the bin midpoint. Small groups make frequencies uncertain.

Which worked group overstates positive frequency and by how much? Explain the steps or evidence behind your response.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Four-step worked evidence for Week 14: Scores and Calibration Intuition

STEP	EVIDENCE AND REASONING
1	Each score estimates the probability that the selected label is correct. Ten predictions have such scores near 0.8.
2	Only six are correct, so observed correctness is 6 ÷ 10 = 60%.
3	The score ranking may still help, but 0.8 should not be stated as an 80% guarantee.
4	Use bins across many cases and design an uncertain zone requiring human review.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

STEP 1 Each **score** estimates the probability that the selected label is correct. Ten **prediction**s have such **score**s near 0.8.

STEP 2 Only six are correct, so observed correctness is 6 ÷ 10 = **60%**.

STEP 3 The **score** ranking may still help, but 0.8 should not be stated as an **80%** guarantee.

STEP 4 Use bins across many cases and design an uncertain zone requiring human review.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a **score**.

Recalculate, retrace, or restate the **evidence** independently:

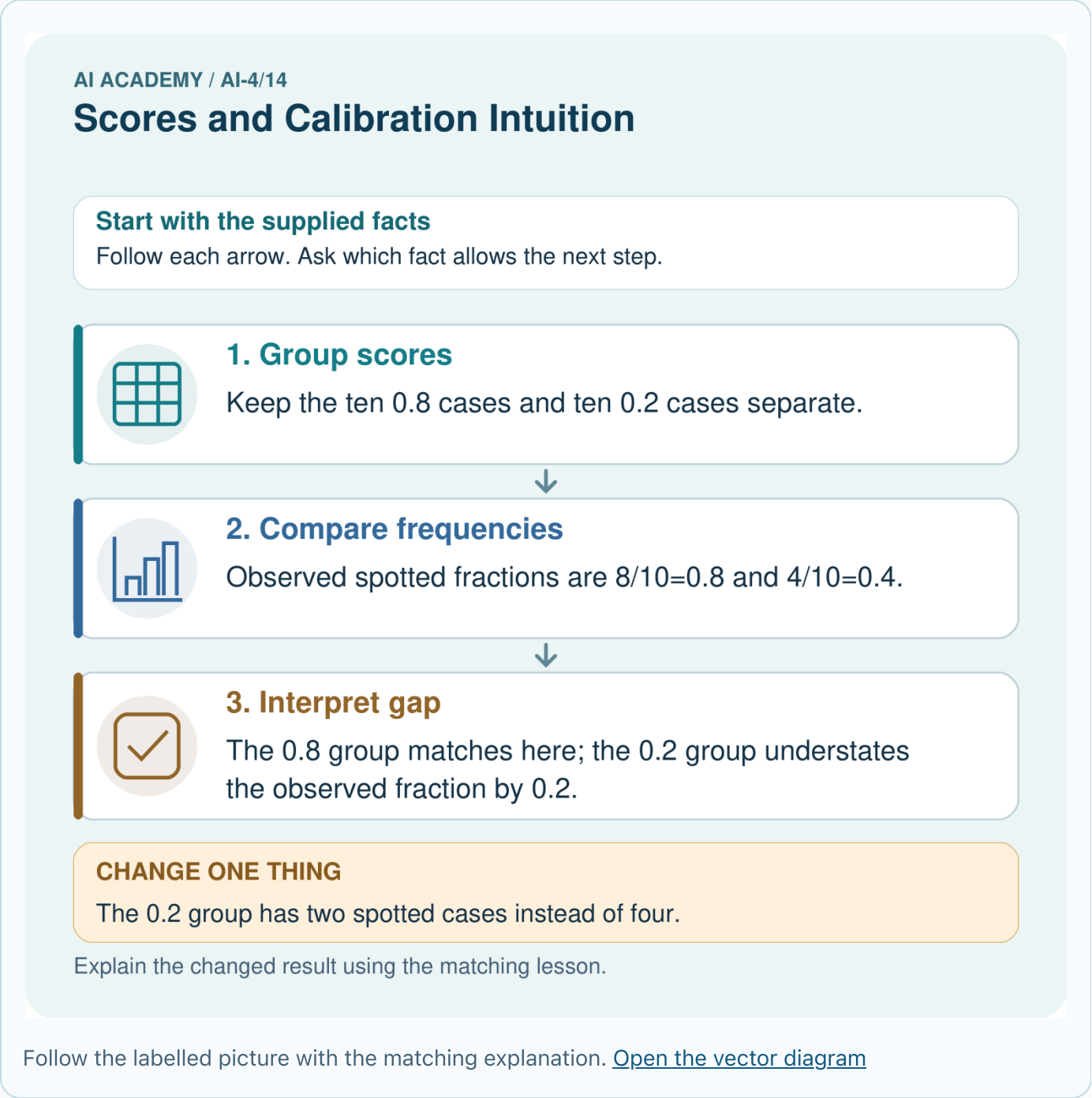
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Scores and Calibration Intuition

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Repair probability-event confusion and test-set reuse

> Name what was forecast and what was observed.

Current event-definition clarification

A score0.2for the positive class does not mean20% of chosen negative predictions should be correct. For example, if a negative label were chosen at threshold0.5, compare correctness with the chosen-label event instead. State that event and policy explicitly. A gap found in one small bin does not justify changing predictions on those same final-test records and calling the resulting fit an independent success.

Inspect sample and final-test limitations

- Bin boundaries, group sizes and dataset coverage affect the picture. Do not fit adjustments on the final test or claim certainty from a small bin.

  Real distributions can change, and a calibrated model on one population may not remain so elsewhere.

What is wrong with calibrating and reporting success on the same final-test records?

Challenge the individual guarantee

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE A **score** of 0.9 means the **prediction** has exactly a 90% chance of being correct.

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A score of 0.9 means the prediction has exactly a 90% chance of being correct. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

Practise the group discrepancy with less help

> A matched frequency is different from useful ranking.

Try W2 with the original two groups

Use workbook W2 for the connected example "**Calibration** as a group property". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Separate calibration, ranking and adjustment data

If every case gets .5 and exactly half are positive, overall **calibration** can look good while the model cannot rank easy and difficult cases. Conversely, useful ranking can coexist with poor probabilities.

Report discrimination and **calibration** separately. Fit any **calibration** adjustment on development data reserved for that purpose, then evaluate on separate data.

Explain a claim with its source and denominator

- Use the worked case for **Calibration** as a group property. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured probability limit

- Does a calibrated 0.8 **score** mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the new independent groups

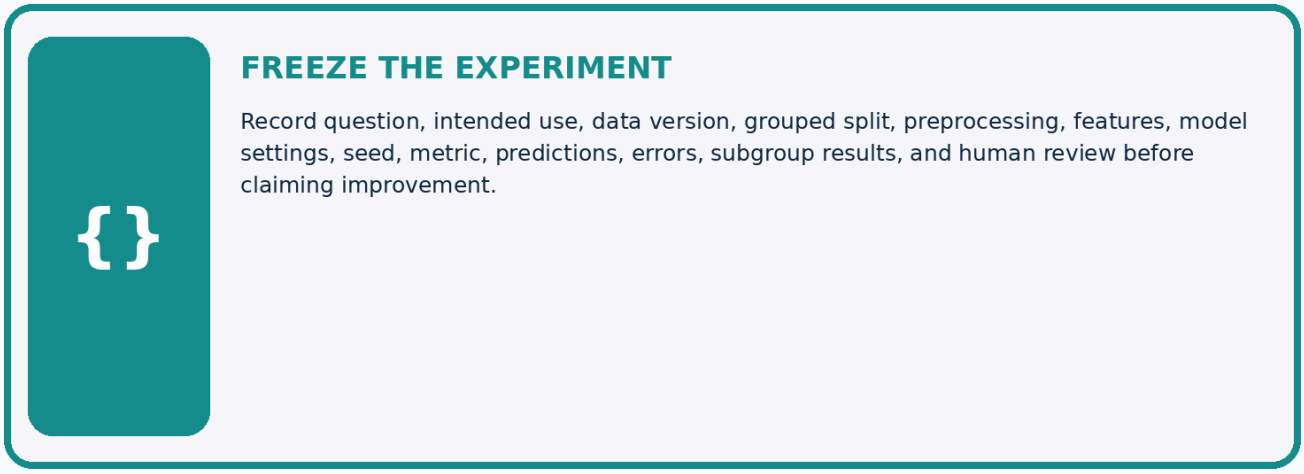
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Locate the discrepancy

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run Reliability Lab with every promised record supplied

> Use observed positive outcomes consistently across all three bins.

Complete thirty fictional score/outcome cards

Case ID	Supplied p(positive)	Observed positive? 1=yes,0=no
L01	0.2	1
L02	0.2	1
L03	0.2	0

Case ID	Supplied p(positive)	Observed positive? 1=yes,0=no
L04	0.2	0
L05	0.2	0
L06	0.2	0
L07	0.2	0
L08	0.2	0
L09	0.2	0
L10	0.2	0
M01	0.5	1
M02	0.5	1
M03	0.5	1
M04	0.5	1
M05	0.5	1
M06	0.5	1
M07	0.5	1
M08	0.5	0
M09	0.5	0
M10	0.5	0
H01	0.8	1
H02	0.8	1
H03	0.8	1
H04	0.8	1

Case ID	Supplied p(positive)	Observed positive? 1=yes,0=no
H05	0.8	1
H06	0.8	1
H07	0.8	0
H08	0.8	0
H09	0.8	0
H10	0.8	0

Declare endpoints and trace counts

Declared bin	Membership rule	Current mean / positive frequency
Low	0<=p<0.35	10cases;mean0.2;2/10=0.2
Middle	0.35<=p<0.65	10cases;mean0.5;7/10=0.7;understates by0.2
High	0.65<=p<=1	10cases;mean0.8;6/10=0.6;overstates by0.2

Normal, exact-endpoint, missing and changed-population probes

Complete probe	Expected bounded check
Normal thirty-record fixture	Low(0.2,0.2),middle(0.5,0.7),high(0.8,0.6); all denominators10.
Boundary p=0.35,observed positive1	Belongs to middle under declared endpoints; this is a separate probe, not an extra main-fixture row.
Missing group with0cases	No observed rate; do not plot0/0as zero. Missing reference outcome also cannot be treated as observed negative.

Complete probe	Expected bounded check
Changed population:10outdoor cases p=0.8with3positives	Observed0.3,gap0.5in this separate tiny sample; original indoor-like evidence cannot certify this setting.

Sort, calculate and interpret

1. Reliability Lab: sort all thirty supplied records under the declared endpoints, count positives and compute the mean probability in each bin. Plot mean score horizontally and observed positive fraction vertically, annotate count10beside each point, and compare with the equality line. Predict before running the optional complete calculation.

   Explain the middle understatement and high overstatement as sample discrepancies, not individual truths. No calibration adjustment or classifier is trained here. Keep the new0.3/0.7independent groups for an uncoached attempt.

Optional complete binning program

```
# Fictional score p(positive), observed binary outcome. Bins declared in advance.
rows = [(f'L{i:02}', 0.2, int(i <= 2)) for i in range(1, 11)]
rows += [(f'M{i:02}', 0.5, int(i <= 7)) for i in range(1, 11)]
rows += [(f'H{i:02}', 0.8, int(i <= 6)) for i in range(1, 11)]
bins = [('low', 0.0, 0.35), ('middle', 0.35, 0.65), ('high', 0.65, 1.0)]
for name, left, right in bins:
    selected = [(p, y) for item_id, p, y in rows
                if left <= p and (p < right or right == 1.0 and p <= right)]
    if not selected:
        print(name, 'no observations')
        continue
    mean_score = sum(p for p, y in selected) / len(selected)
    positives = sum(y for p, y in selected)
    frequency = positives / len(selected)
    print(name, len(selected), round(mean_score, 3), positives, round(frequency, 3))
# These groups describe p(positive), not the correctness rate of chosen labels.
# No calibrator is fitted, and no population guarantee follows from ten cases/bin.
```

Keep evidence and actual execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Missing
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Shifted
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

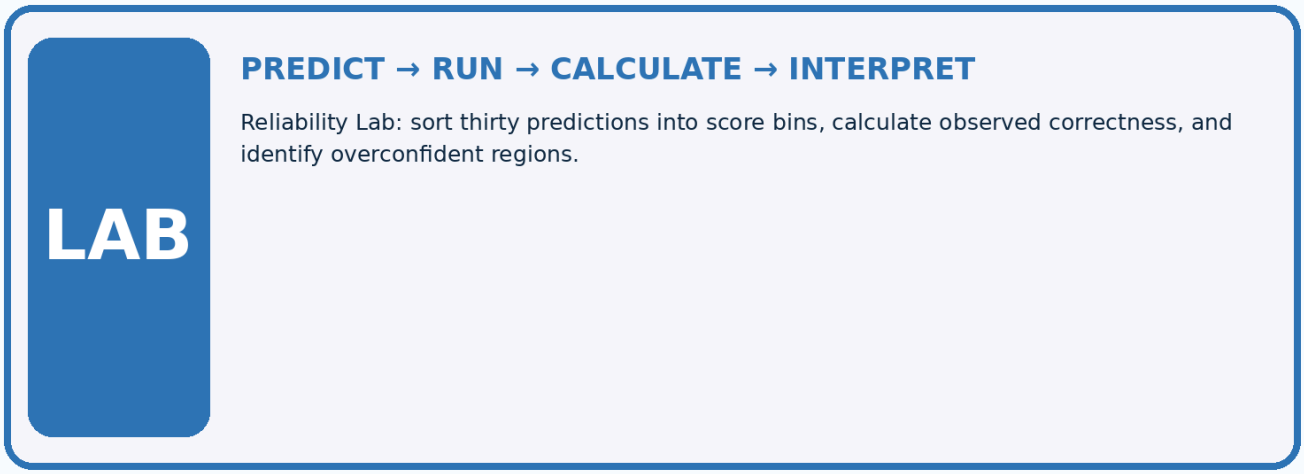
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. **Reliability** Lab: sort thirty supplied p(positive) **prediction**s into declared **score** bins, calculate observed positive frequency and identify sample discrepancies. Keep this distinct from chosen-label correctness.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

Try alone: interpret two new probability groups

> State the score’s event, both points and the direction of each sample gap.

Complete independent facts and scope

Twenty cases all predict .3 with eight positives; ten all predict .7 with seven positives. Give plot points and interpret.

In this selected task the scores mean p(positive). Positive outcomes are the supplied counts, not counts of correct chosen labels. Plot group mean probability on x and observed positive frequency on y; show the two denominators and state the limited evidence. No per-case prediction decision rule, global calibration test or future guarantee is supplied.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply independently

Week 14 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Among 20 **prediction**s with **score** 0.8, 12 are correct. What is observed accuracy in this group and how does it compare with 0.8?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Add an uncertainty band to the Plant Classifier and define what the interface says when **score**s fall inside it.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Twenty cases receive **score** .7 and fourteen are correct. Explain what the observed rate suggests and what it does not establish about another **score** group.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep score meaning beside the project's review policy

> Do not turn a toy reliability check into a validated uncertainty claim.

Project score record and limits

THIS WEEK'S CHECKPOINT Add an uncertainty band to the Plant Classifier and define what the interface says when **score**s fall inside it.

Define the score event and reference label first. Save score-bin rules, counts, mean probabilities, observed frequencies and the data/version/condition context. A proposed review band is a development policy requiring its own justification; low positive probability need not mean high uncertainty, and a score near0.5does not by itself measure every uncertainty.

Do not fit a calibrator or review threshold on final-test outcomes and then report an independent final success. Record calibration adjustment as not performed in this lesson.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence, uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

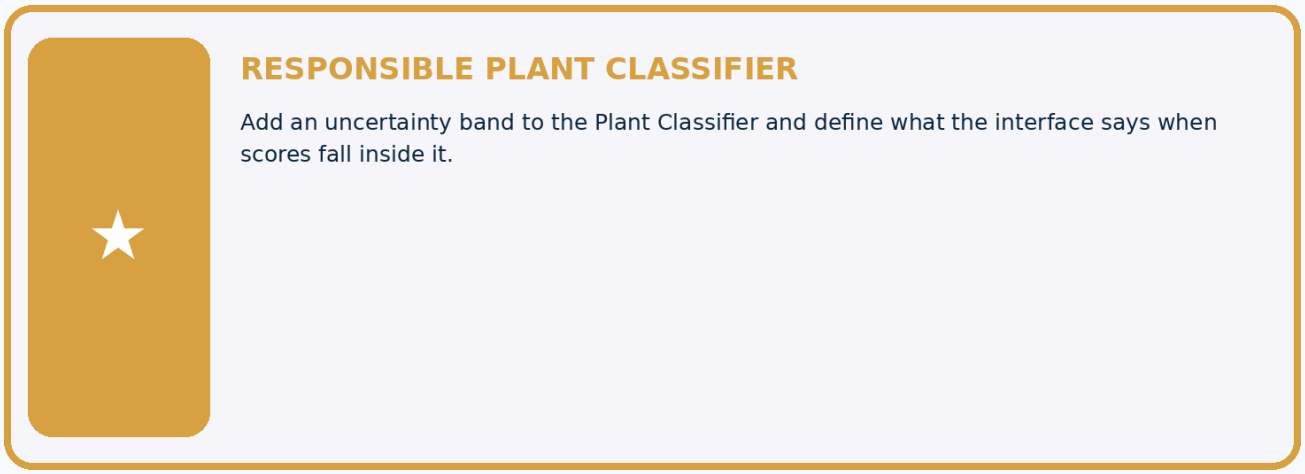
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish by naming the event, the bin and the denominator

> Explain why a matched group does not make every individual outcome certain.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Interpret **calibration** using groups of forecasts

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
score	model value used for ranking or confidence	
calibration	scores align with observed frequencies	
bin	group of predictions with similar scores	
reliability	agreement between score and outcome rate	

Closer-explanation terms

Word	Meaning
Calibration	Agreement between predicted probabilities and outcome frequencies across comparable groups
Bin	A declared interval grouping predictions

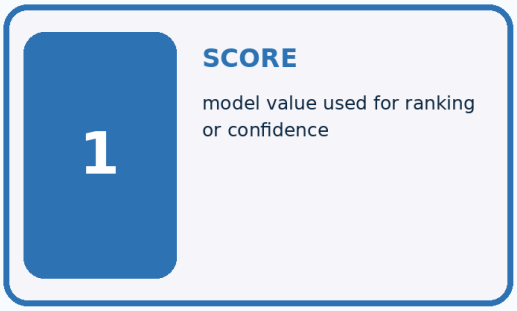
Word	Meaning
Observed frequency	Positive outcomes divided by cases in a group
Reliability diagram	A plot comparing group average prediction with observed frequency

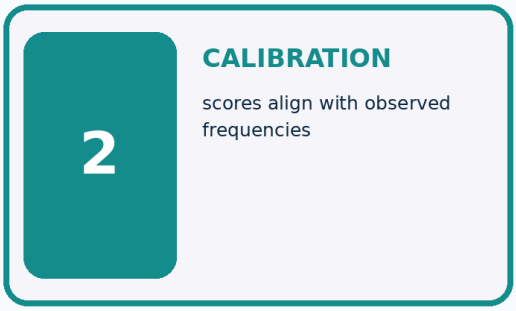
▼ Earlier wording, alternatives and picture transcripts

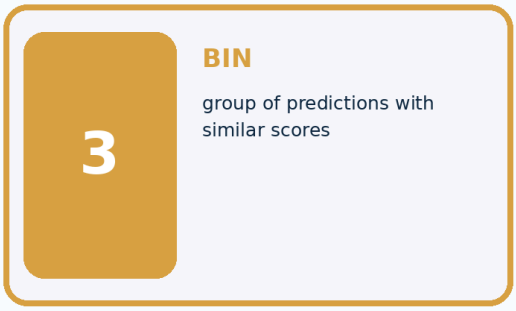
VOCABULARY

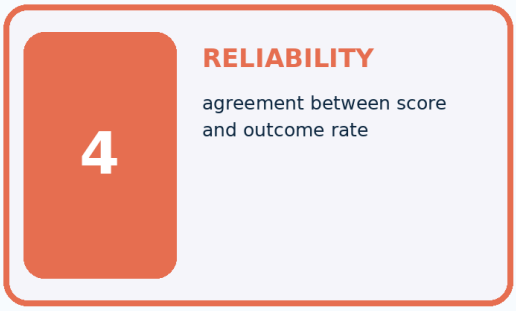
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A **score** of 0.9 means the **prediction** has exactly a 90% chance of being correct.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **score**, **calibration**, bin, **reliability**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A **score** of 0.9 means the **prediction** has exactly a 90% chance of being correct.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

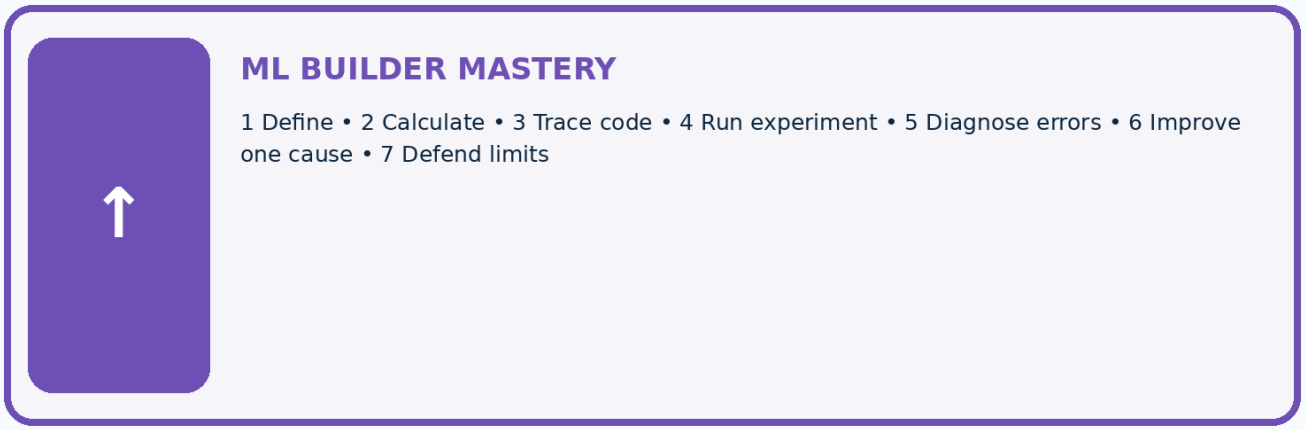
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the fresh two-group positive-probability interpretation task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

- Independent **evidence** and defense:

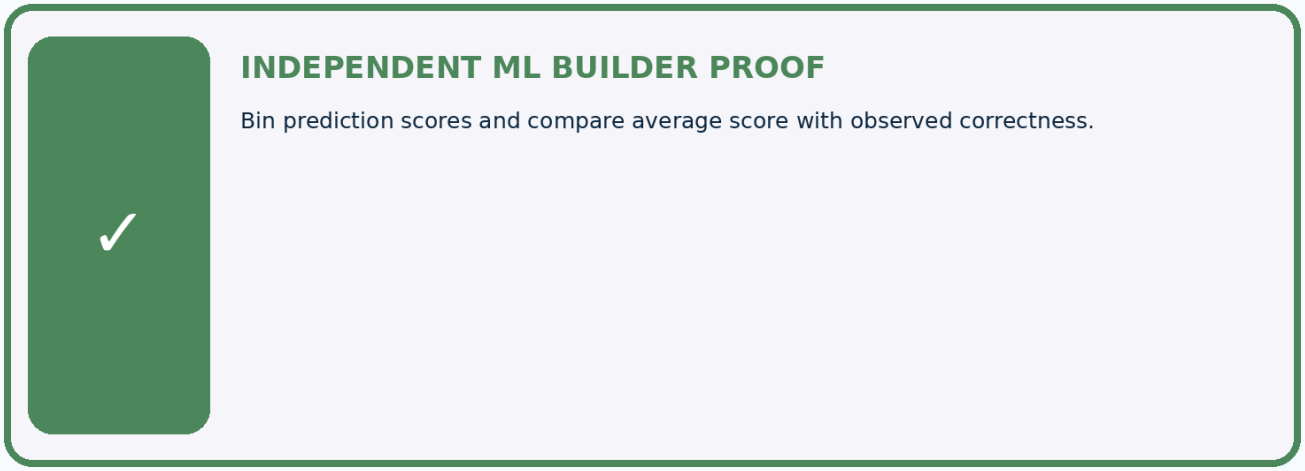
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Bin **prediction score**s and compare average **score** with observed correctness.

NEXT BRIDGE Carry the experiment **evidence** into: Confusion Matrix.

▼ **Extra practice · Earlier complete case**

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Ten cases receive probability .8 for spotted; eight are spotted.

Interpret **calibration** and its limits.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

8/10 matches .8 in this tiny bin. It illustrates **calibration**, not proof of global **calibration** or individual certainty; test more bins and data.

Evidence or stage	Value or result
Spotted among score-0.8 cases	8/10

▼ Earlier wording, alternatives and picture transcripts

Week 14 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Compare a score bin with observed frequency

Ten cases receive probability .8 for spotted; eight are spotted.

Interpret **calibration** and its limits.

8/10 matches .8 in this tiny bin. It illustrates **calibration**, not proof of global **calibration** or individual certainty; test more bins and data.

▼ **Reference · Earlier route checklist**

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Interpret calibration using groups of forecasts	Explain the deciding step, show a trace or test, and state one limit.
Compare average predicted probability with observed frequency	Explain the deciding step, show a trace or test, and state one limit.
Distinguish calibration from individual correctness	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: Treat **calibration** as a comparison across groups of probability **score**s.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Treat **calibration** as a comparison across groups of probability **score**s.
- Which labelled step matches this detail? The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.
- 1. Group **score**s
- 2. Compare frequencies
- 3. Interpret gap

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Interpret gap The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. Read the picture in order.

The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The 0.2 group has two spotted cases instead of four. Which conclusion is supported by the supplied facts?
- 1. Its observed fraction becomes 0.2 and matches the supplied **score** in this group.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Its observed fraction becomes 0.2 and matches the supplied **score** in this group. **Calibration** compares stated probabilities with group outcome frequencies.

Repair and keep the record

Review Train a Classifier (ai-4/13). Goal: Connect **training**-only scaling and neighbour **prediction** in one pipeline. Short practice: Which pipeline record would help another team reproduce the result?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Which **input** distinguishes the original and changed tree?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Leakage (ai-4/8). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many independently new plant identities are in the original test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

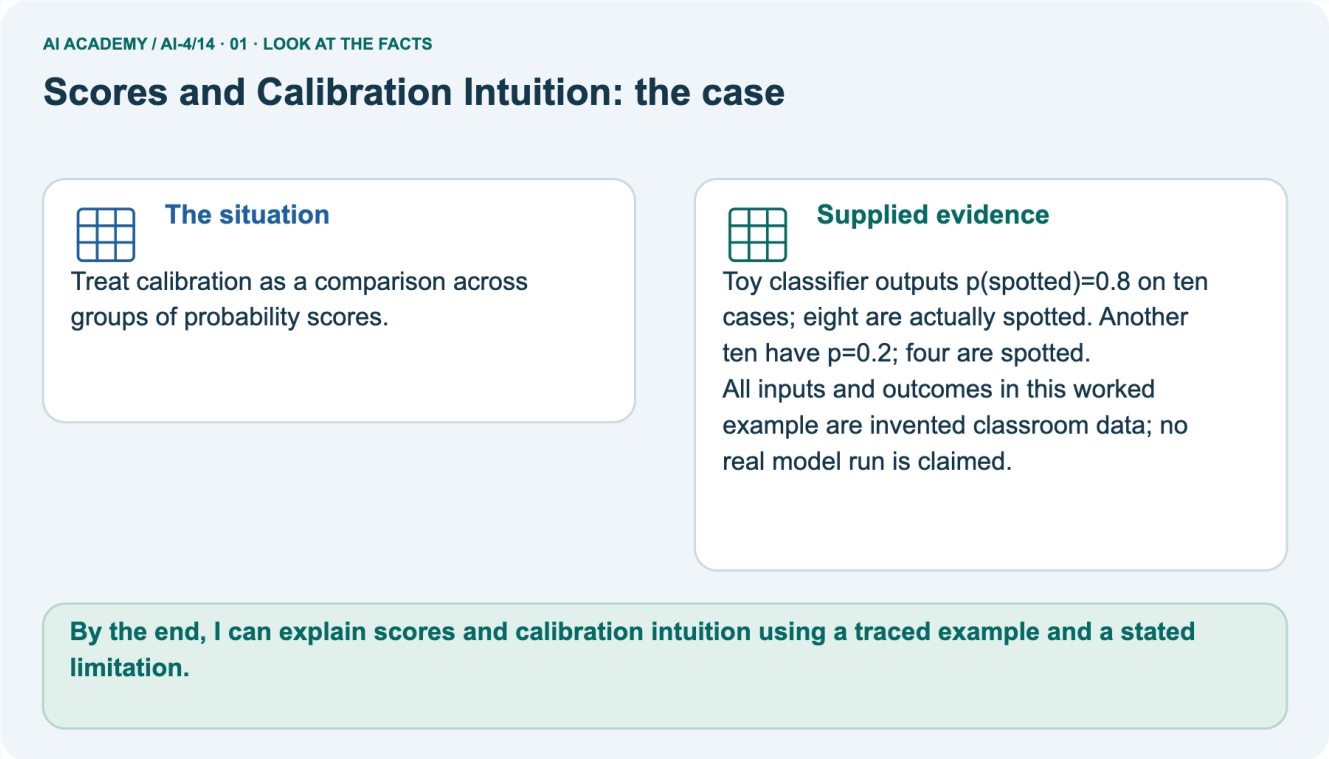
▼ Reference · The six-picture journey

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Scores and Calibration Intuition: the case



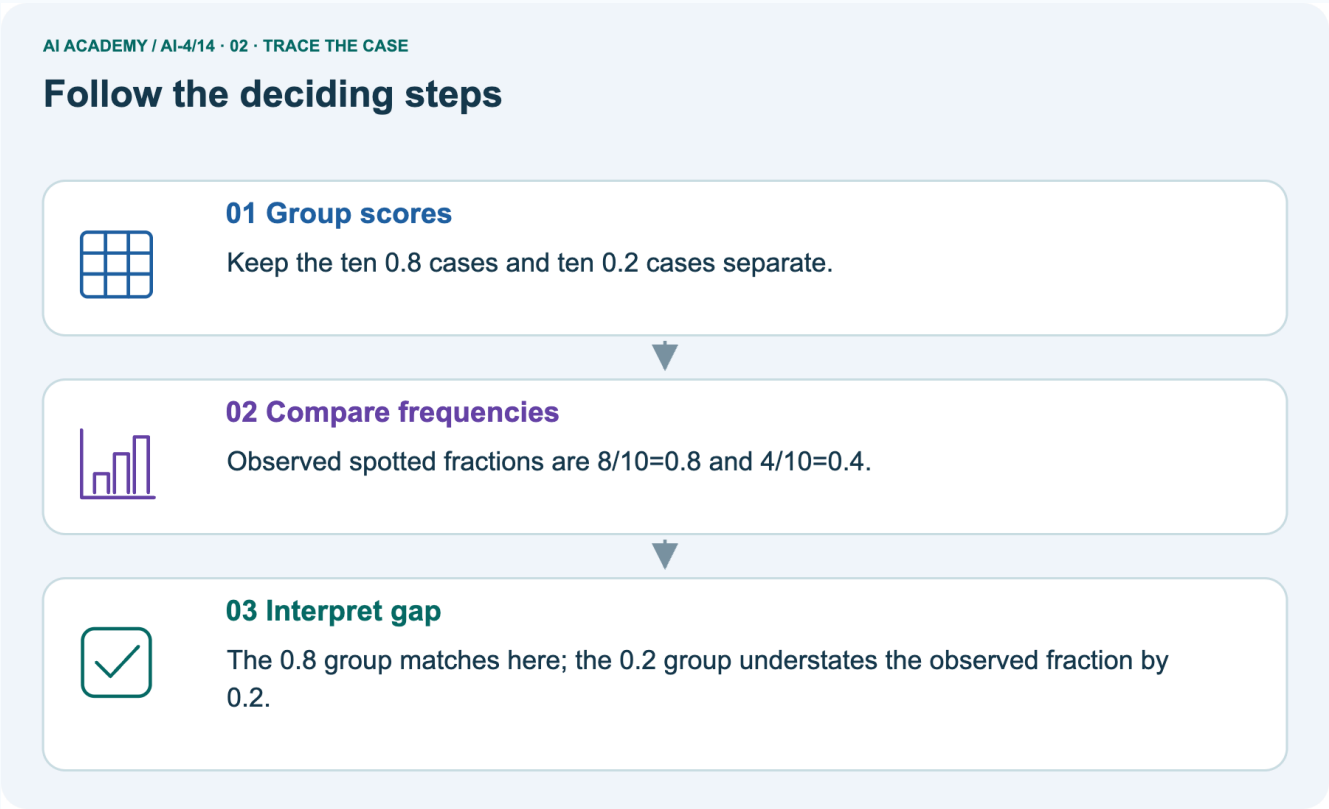
Follow the labelled picture with the matching explanation. Open the vector diagram

Treat **calibration** as a comparison across groups of probability **score**s. Supplied facts: Toy classifier **output**s p(spotted)=0.8 on ten cases; eight are actually spotted. Another ten have p=0.2; four are spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Group **score**s: Keep the ten 0.8 cases and ten 0.2 cases separate. → Compare frequencies: Observed spotted fractions are 8/10=0.8 and 4/10=0.4. → Interpret gap: The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

Point to each numbered stage and explain which recorded **input** it uses.

03. Four-step worked evidence for Week 14: Scores and Calibration

Intuition

AI ACADEMY / AI-4/14 · 03 · INSPECT THE LESSON MODEL

Four-step worked evidence for Week 14: Scores and Calibration Intuition

Step	Evidence and reasoning
1	Each score estimates the probability that the selected label is correct. Ten predictions have such scores near 0.8.
2	Only six are correct, so observed correctness is 6 ÷ 10 = 60%.
3	The score ranking may still help, but 0.8 should not be stated as an 80% guarantee.
4	Use bins across many cases and design an uncertain zone requiring human review.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

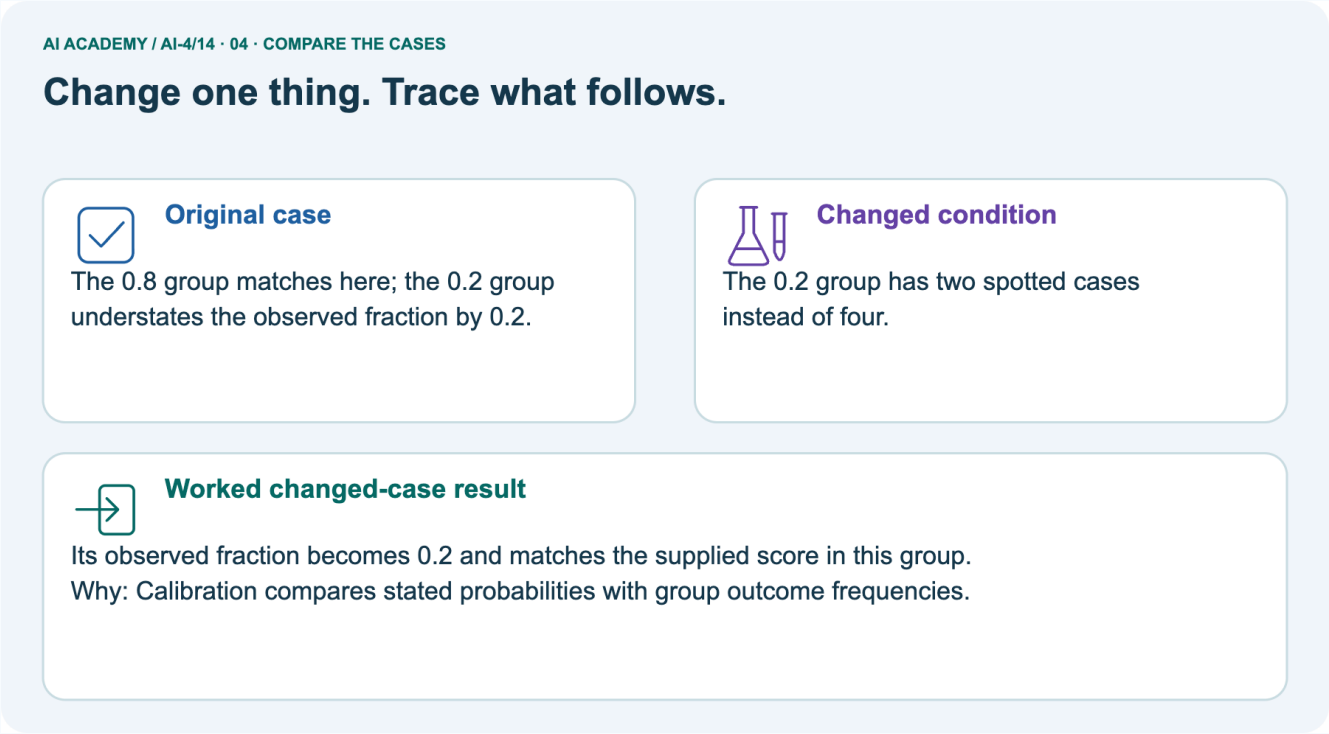
Follow the labelled picture with the matching explanation. Open the vector diagram

Four-step worked **evidence** for Week 14: **Score**s and **Calibration** Intuition. 1 | Each **score** estimates the probability that the selected label is correct.

Ten **prediction**s have such **score**s near 0.8.. 2 | Only six are correct, so observed correctness is 6 ÷ 10 = **60%**.. 3 | The **score** ranking may still help, but 0.8 should not be stated as an **80%** guarantee.. 4 | Use bins across many cases and design an uncertain zone requiring human review.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



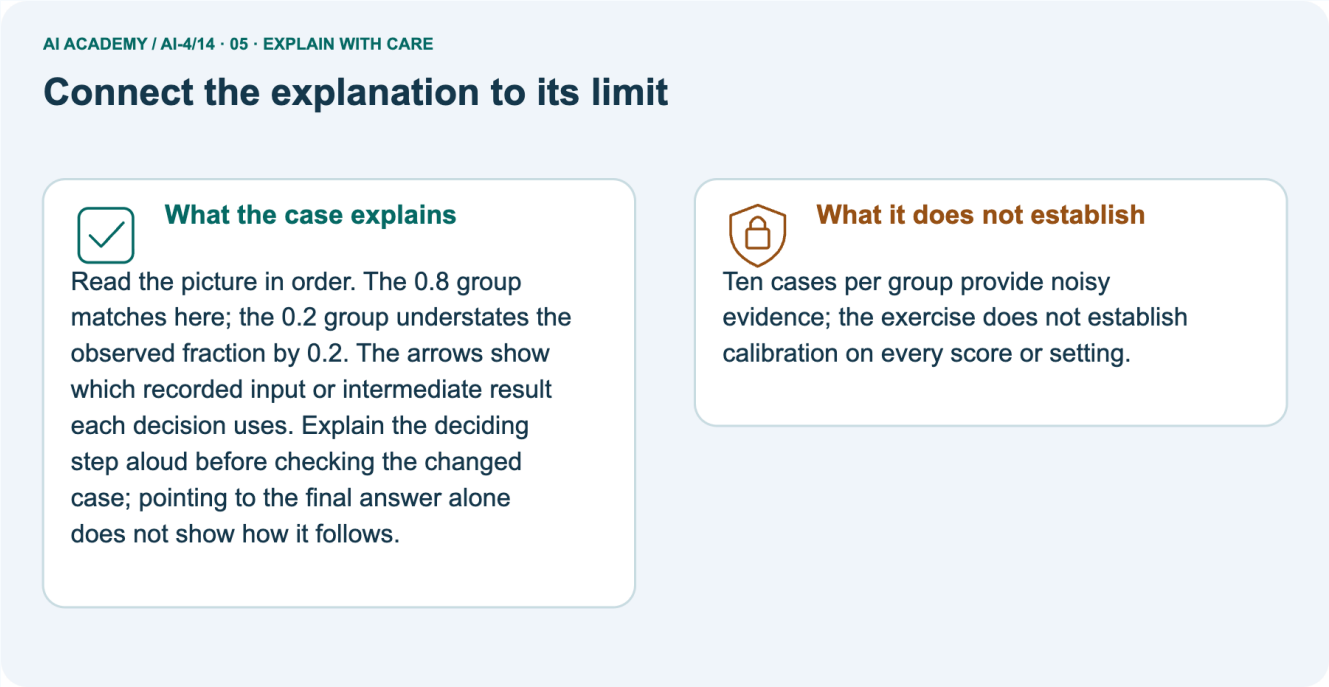
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. Changed condition: The 0.2 group has two spotted cases instead of four.

Expected result: Its observed fraction becomes 0.2 and matches the supplied **score** in this group. Reason: **Calibration** compares stated probabilities with group outcome frequencies.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

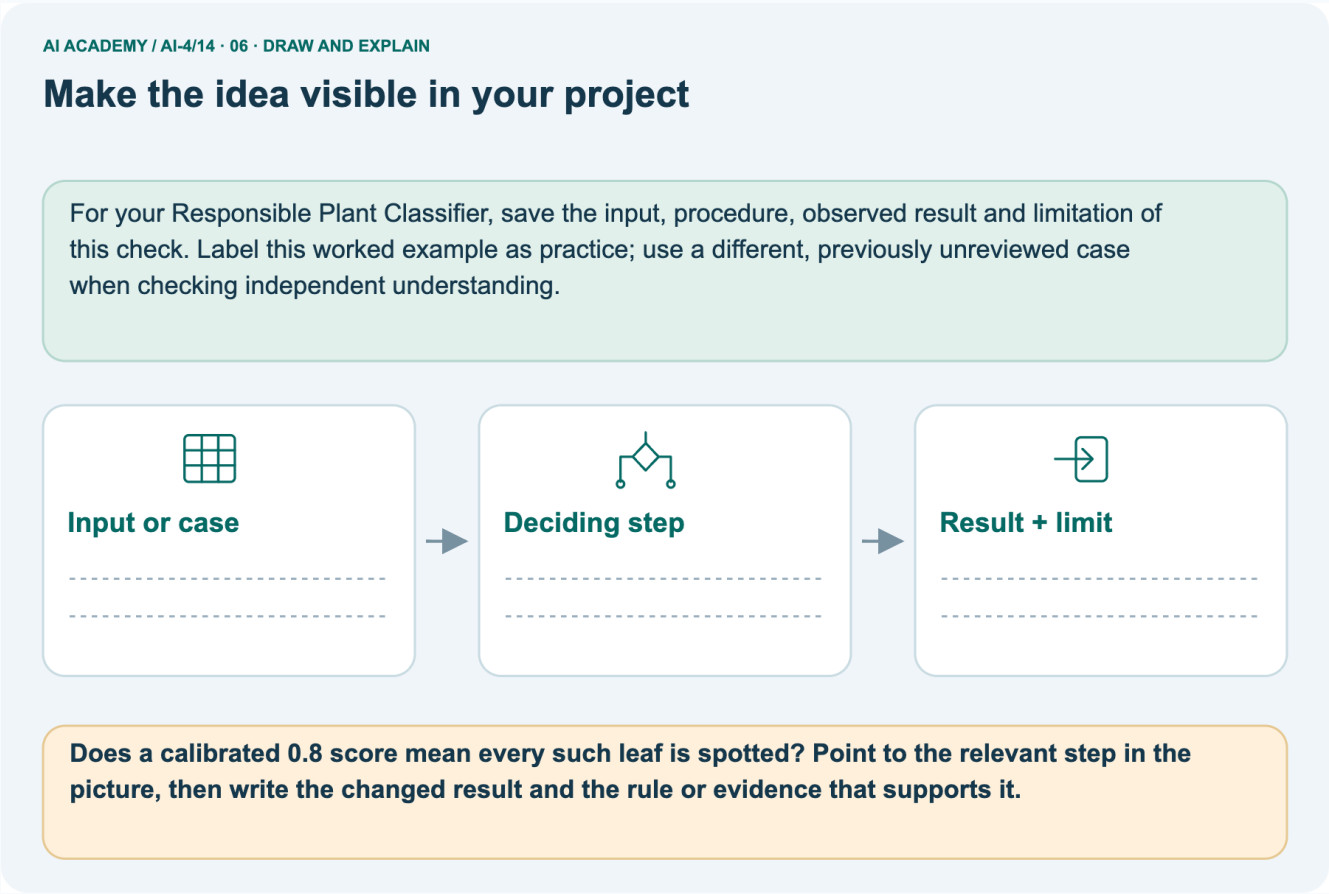
Read the picture in order. The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Ten cases per group provide noisy **evidence**; the exercise does not establish **calibration** on every **score** or setting.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does a calibrated 0.8 **score** mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Scores and Calibration Intuition: the case

02. Follow the deciding steps

03. Four-step worked evidence for Week 14: Scores and Calibration Intuition

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.